

Ebrahim Tahmasebi Boldaji

Naval Architect & CFD Engineer | Hydrodynamics, Simulation & Machine Learning

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Profile

Naval architect and CFD engineer with 7+ years of experience spanning ship design, high-fidelity hydrodynamic simulation, and machine-learning surrogate modelling. Double master's background (Erasmus Mundus SEAS 4.0, grade 109/110) combining classical naval architecture with Industry 4.0 methods. Two first-author journal publications; hands-on shipyard quality-control experience during newbuild construction. Comfortable moving between production environments, classification rules, and research-grade simulation tools. Quick to pick up new tools and stubborn about finishing what I start — evidenced by building four solver types from first principles and carrying two papers from first simulation through to publication.

Technical Skills

CFD & Simulation	Simcenter STAR-CCM+, OpenFOAM, FreSCo+ (in-house RANS, TUHH), ANSYS AQWA; VOF free-surface flows, seakeeping, resistance & propulsion, overset/dynamic meshes, $k-\omega$ SST turbulence modelling, mesh independence & verification studies
Solver Development	Custom FVM, FDM, SPH and LBM solvers implemented from first principles (C++/Python); fluid-structure interaction via STAR-CCM+/ABAQUS coupling
Machine Learning	Surrogate models for CFD (CVAE, POD/ROM, GNN, PINNs); training pipelines, validation, and integration into production solvers
Structures	ABAQUS, finite element programming, inverse FEM for structural health monitoring
Programming & Tools	Python, C++, MATLAB; Linux, Git/GitHub, HPC clusters, LaTeX; SolidWorks, Rhino, AutoCAD, Maxsurf
Domain Knowledge	Ship design (GA, hydrostatics, intact stability, propulsion selection); class & regulatory frameworks (SOLAS, MARPOL, IMO, FSICR ice class, CRS scantling); alternative fuels under CII/EEXI/FuelEU

Professional Experience

M.Sc. Researcher — ML-Assisted CFD Sep 2025 – Feb 2026
Technical University of Hamburg (TUHH), Institute for Fluid Dynamics and Ship Theory (M8) | Hamburg, Germany

- Developed three CVAE-based surrogate models reconstructing velocity fields around a Voith propeller from high-fidelity CFD data, cutting field-prediction cost from full simulations to near-instant inference
- Integrated the trained surrogates directly into the in-house RANS solver FreSCo+ for fast hull drag prediction, interfacing the ML inference layer with the existing solver architecture
- Built fully automated Python pipelines covering simulation execution, data extraction, training, and validation

Quality Control Engineer Sep 2023 – Sep 2024
Shipyards — aluminium newbuild construction

- Performed dimensional checks, plate thickness measurements, and weld inspections during hull assembly of new aluminium vessels, verifying compliance with class drawings and construction standards
- Monitored structural assembly, coordinated with production teams on non-conformities, and maintained inspection records for project documentation

Lecturer Sep 2023 – Aug 2024
Persian Gulf University

- Delivered undergraduate courses in Ship Design and CAD Modelling

Research Assistant Sep 2022 – Aug 2023
Persian Gulf University

- Analysed hydrodynamic performance of heave plates for offshore structures using RANS CFD, validating results against experimental data across multiple operating conditions

M.Sc. Researcher — Fluid-Structure Interaction
Sharif University of Technology

Sep 2019 – Mar 2021

- Developed a coupled STAR-CCM+/ABAQUS framework for FSI problems, handling interface data transfer, mesh deformation, and coupled-solution convergence
- Implemented inverse finite element solvers from first principles in MATLAB (stiffness assembly, boundary conditions, Tikhonov regularisation) for real-time structural health monitoring

Teaching Assistant
Sharif University of Technology & Persian Gulf University

2016 – 2020

- Courses: Ship Structures, Fluid Dynamics, Ship Design, Mechanical Vibrations

Education

Erasmus Mundus Joint Master — Sustainable Ship and Shipping 4.0 (SEAS 4.0)
Technical University of Hamburg (DE) · University of A Coruña (ES) · University of Naples Federico II (IT) | Grade: 109/110

Sep 2024 – Mar 2026

- Thesis: *Machine-Learning Assisted Field Surrogate Models of Complex Ship Propulsion Systems in Hydrodynamic Numerical Simulations*
- Focus: marine CFD, ship design & seakeeping, arctic technology and ice-class design, alternative fuels, Industry 4.0 enabling technologies

M.Sc. Naval Architecture
Sharif University of Technology | GPA: 3.87/4.0

Sep 2018 – Mar 2021

- Thesis: *Structural Health Monitoring of Marine Structures with Inverse Finite Element Method*

B.Sc. Naval Architecture and Marine Engineering
Persian Gulf University | GPA: 3.70/4.0 — Ranked 1st in cohort

Sep 2014 – Jul 2018

- Thesis: *Investigation of Hydrodynamic Behaviour of Planing Craft using STAR-CCM+*

Publications

- E. Tahmasebi Boldaji et al., *Hybrid Negative-to-Positive Deadrise Hulls: A Performance Assessment for High-Speed Planing Vessels*, Ocean Engineering, 2026. [\[link\]](#)
- E. Tahmasebi Boldaji et al., *The Effects of Hydrodynamic Load on the Heave and Pitch Coupling Oscillations of a Plate*, Journal of Marine Science and Technology, 2024. [\[link\]](#)

Awards & Languages

Awards	Erasmus Mundus Scholarship (2024) — fully funded, awarded on academic excellence; 1 st rank, B.Sc. Naval Architecture cohort (2014–2018)
Languages	English — full professional proficiency (C1); German — beginner, actively improving; Persian — native

References

Prof. Dr.-Ing. Thomas Rung Hamburg University of Technology thomas.rung@tuhh.de	Dr.-Ing. Georgios Bletsos Hamburg University of Technology george.bletsos@tuhh.de	Prof. Ermina Begovic University of Naples Federico II begovic@unina.it
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